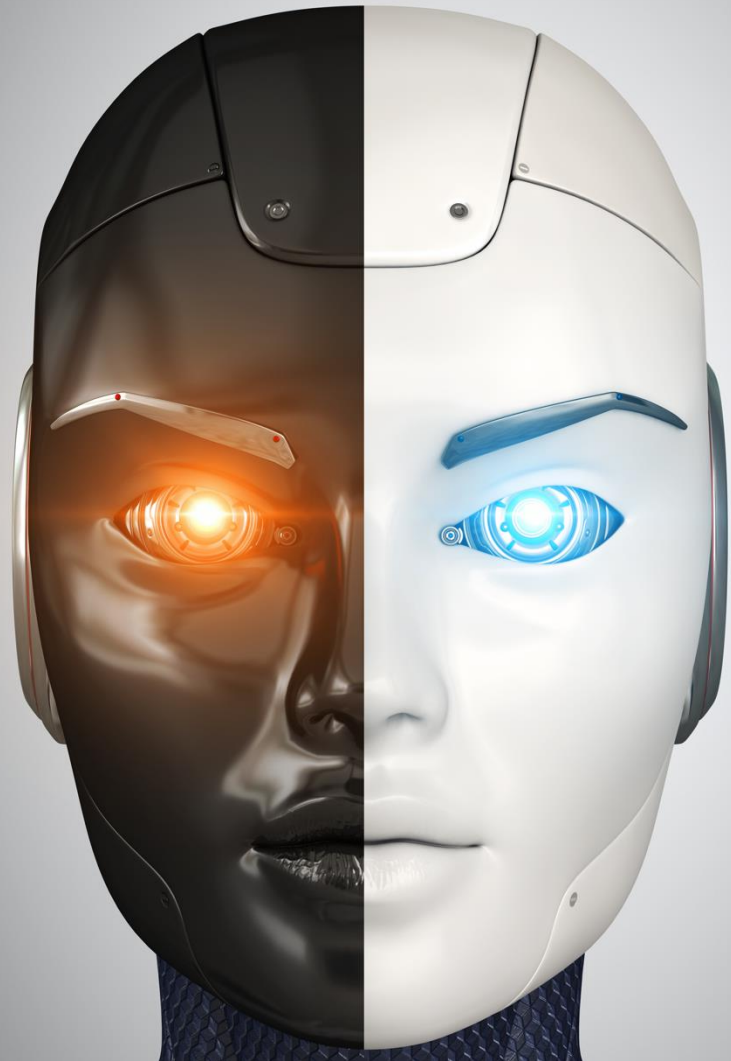


# About Artificial Intelligence



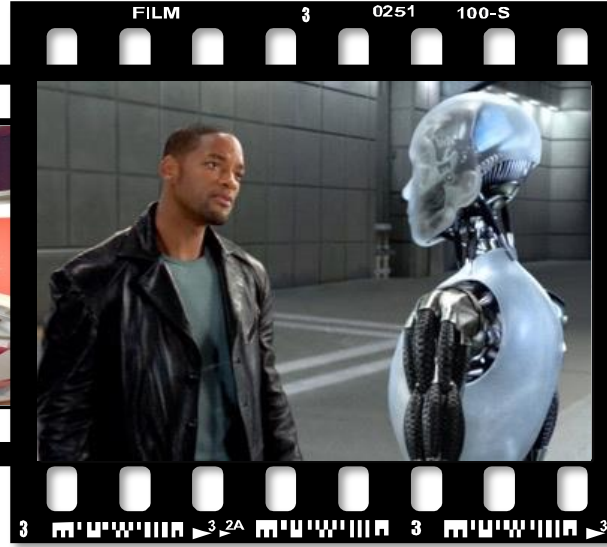
What is your  
definition of  
**Artificial  
Intelligence?**



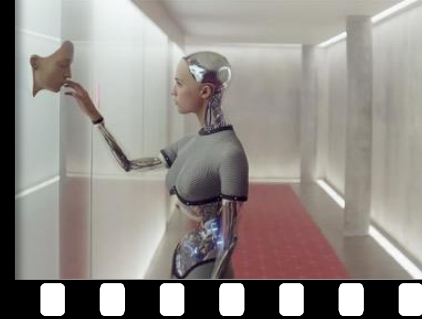
# ARTIFICIAL INTELLIGENCE

## Perception *TODAY*

I, Robot



Ex-Machina



The Matrix

Her

Highly influenced by media and  
entertainment industry

# Here is a breakdown of AI...

If I say, **Sky**, you would likely say....

Blue



If I say, **grass**, you would likely say....

Green

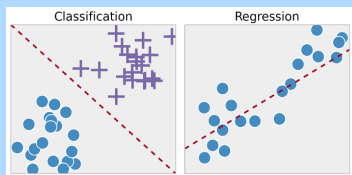
If I say, **AI**, I encourage you to say....

Machine Learning

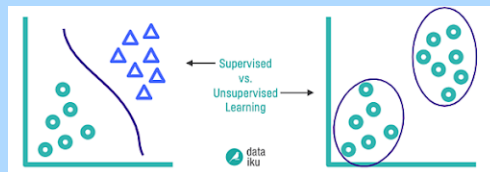
If I say, **machine learning**, I'd like to think...

Prediction

## 1) Supervised learning



## 2) Unsupervised learning



## 3) Reinforcement learning



...and what comprises **algorithms'** favorite food???....

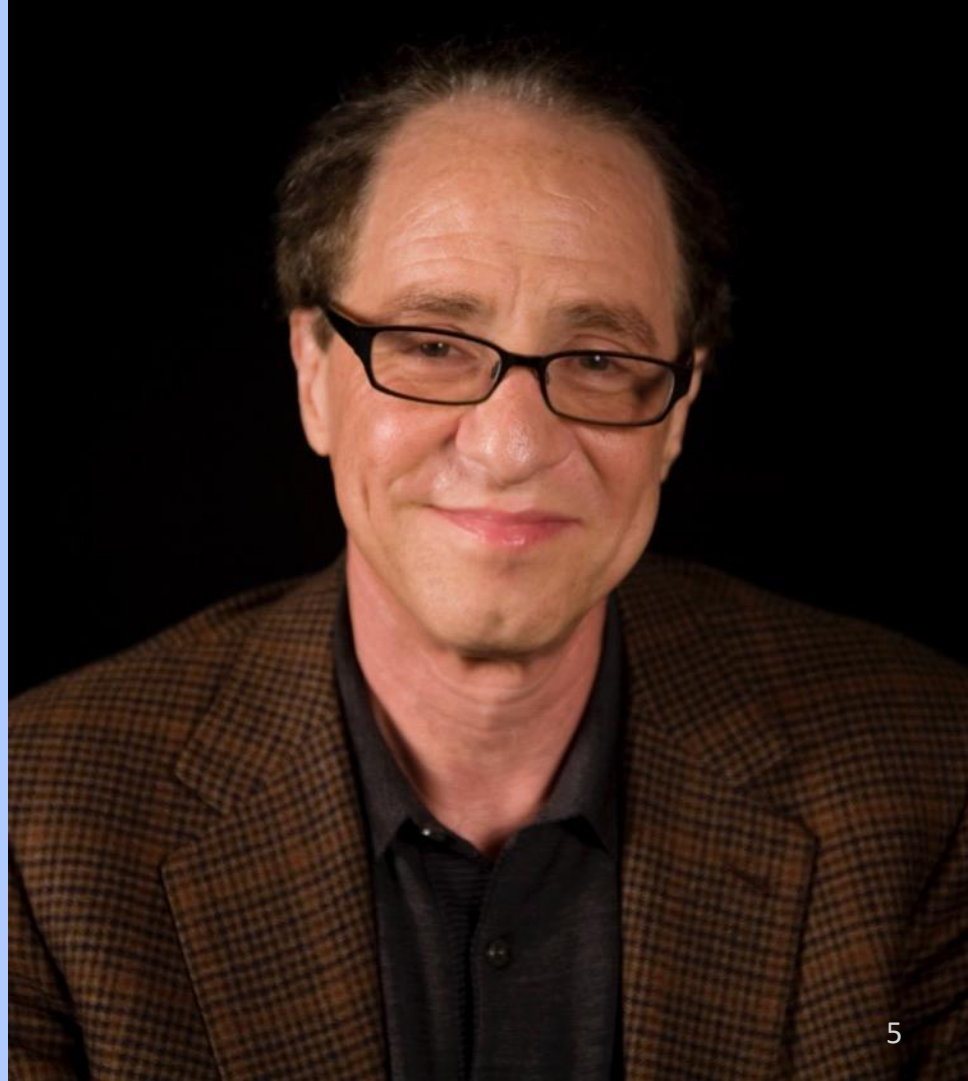
Data

“Our technology, our machines are **part of our humanity.**

We create them to extend ourselves and that is what is unique about human beings”

**Ray Kurzweil**

Futurist





# January 2011: Jeopardy!

Ken, Watson and Brad



Training Watson



# AI applications are transforming every industry



## Government

Campaign Content and Planning, Citizen Experience, Public Security, Policy Planning Support



## Finance

High Frequency Trading, Risk Modeling, Equity Research, Asset Management, Underwriting, Investment Planning, Security



## Agriculture

UAV / Satellite Crop Field Analysis, Disease Recognition, Comprehensive Strategic Crop Planning



## Energy

Strategic Oil Drilling, Risk Minimization, Geological Analysis, Demand Prediction, Adjustment of Resource Generation



## Healthcare

Personalized Healthcare, Diagnostic Tools, Integrated Wellness and Health Systems, Behavior Tracking, Security



## Education

Personalized Education, Learning Content Indexing-to-Skill & Search, Custom Teaching Methods, Smart View Devices



## Science

Data Analysis, Experiments, Predictive Modeling, Theorem Proving, Deductive Reasoning, Experiment Planning



## Business Solutions

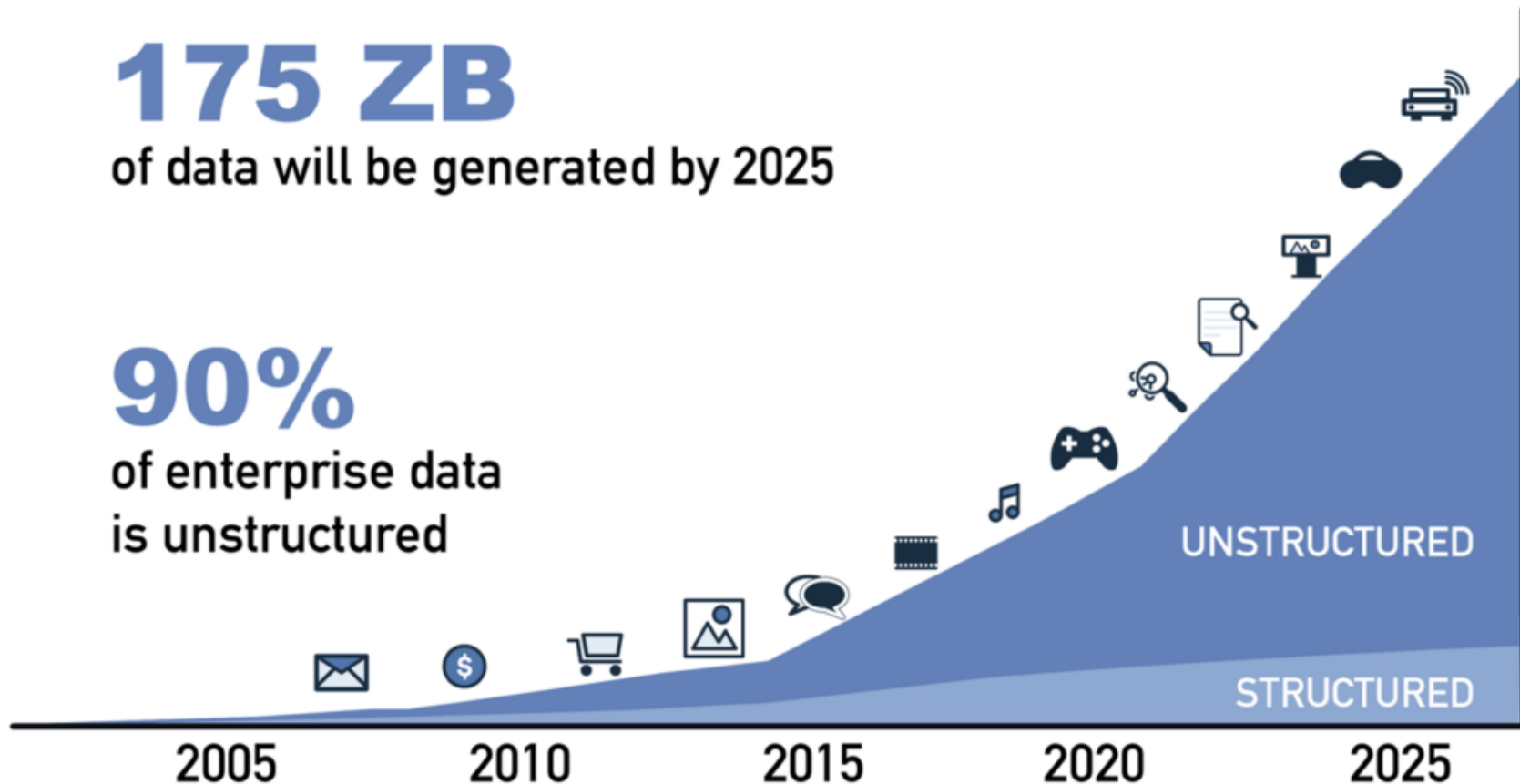
Interactive Chatbots that Learn from Experience with Customers, Regulatory Support, Prediction, Marketing

# 175 ZB

of data will be generated by 2025

## 90%

of enterprise data  
is unstructured



SOURCE: IDC



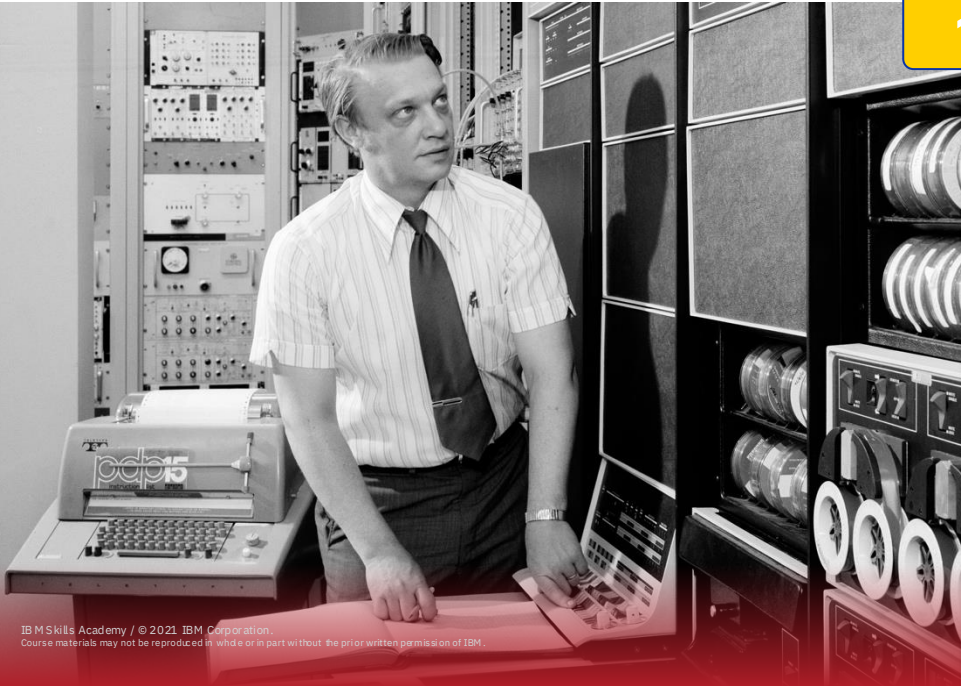
# Traditional systems and Cognitive systems

Traditional systems

**Deterministic**  
**Static**

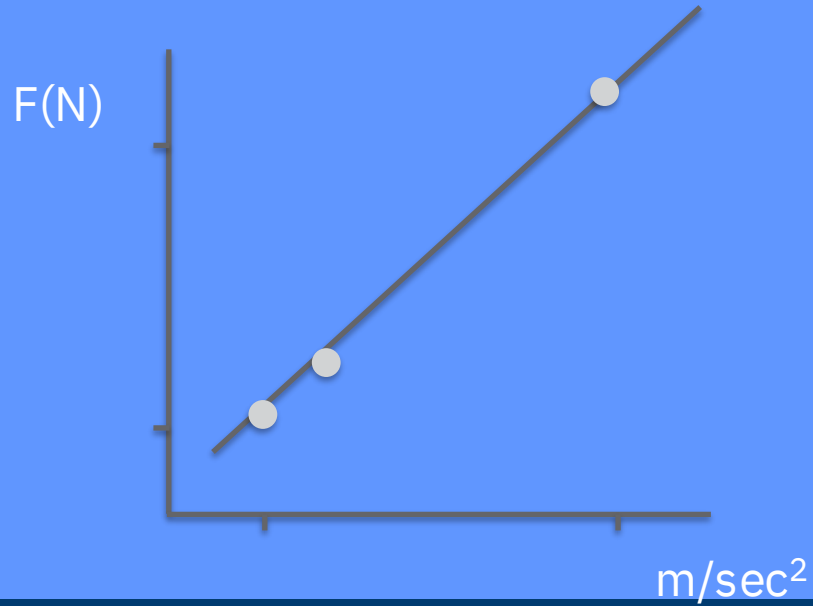
Cognitive systems

**Probabilistic**  
**Self-improving**



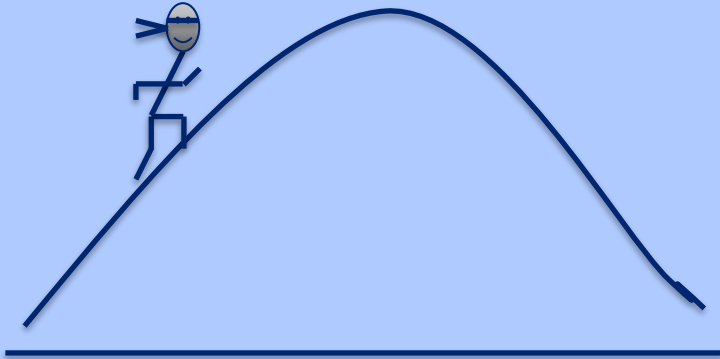
Traditional

# Deterministic Systems



$$F=ma$$
$$C=2Kg$$

# Machine learning is blind

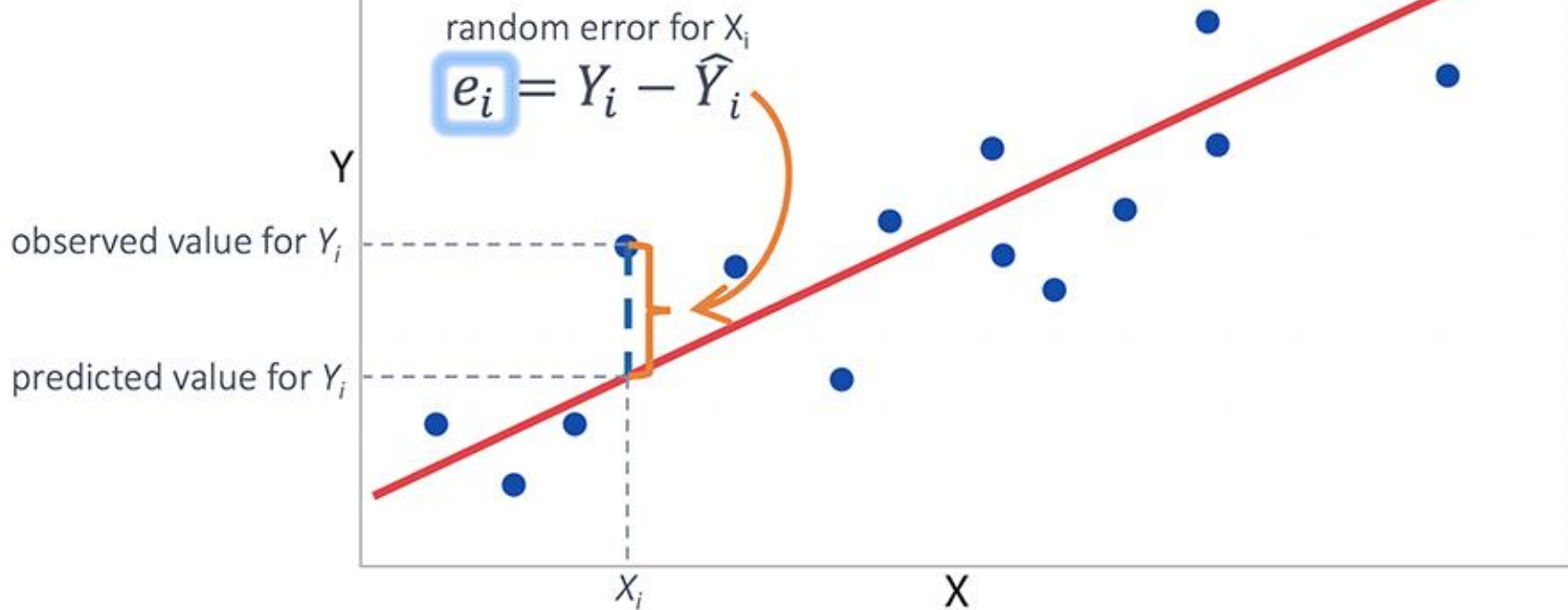


Climb to the top of the hill  
Under two conditions:

- a) Must do it blindfolded
- b) In as minimum steps as possible

## Method of Least Squares

$$\sum e_i^2 = \sum (Y_i - \hat{Y}_i)^2$$



# Do the Picaso

# Predicting your grades using neural nets

| X | hrs study | hrs sleep | Grade | y         |
|---|-----------|-----------|-------|-----------|
|   |           |           |       |           |
|   | 7         | 5         | 78    |           |
|   | 6         | 8         | 93    |           |
|   | 8         | 2         | 67    |           |
|   | 5         | 5         | ?     | $\hat{y}$ |

Supervised learning

This is a regression problem

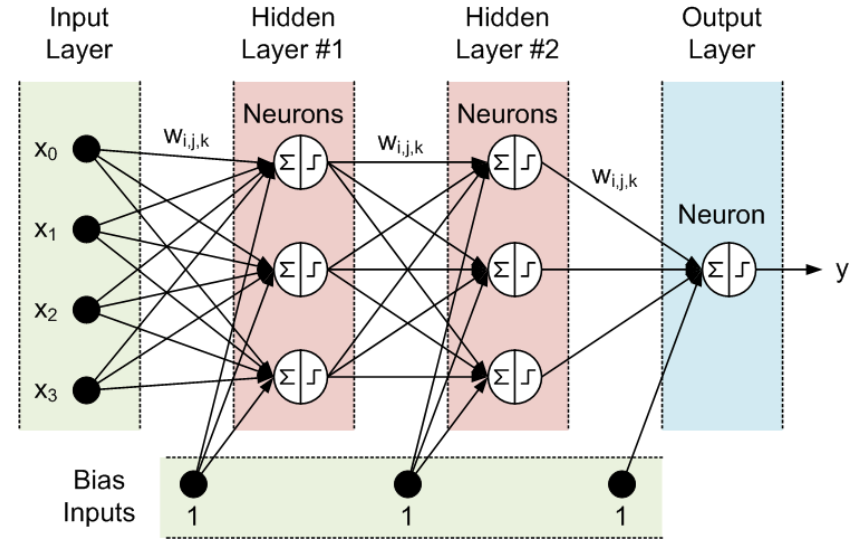
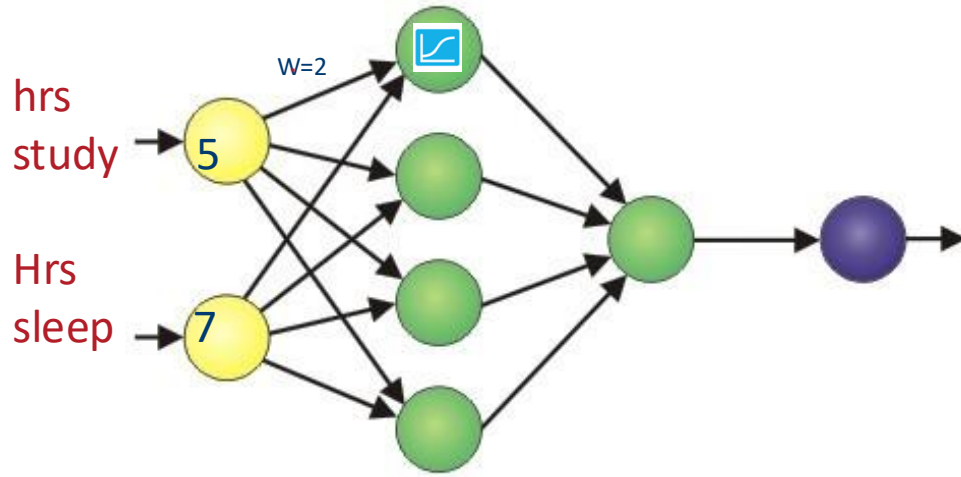
Not a classification problem

$$X_{\text{norm}} = x / \max(x)$$

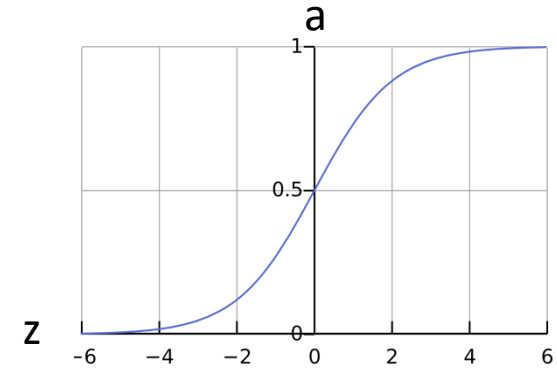
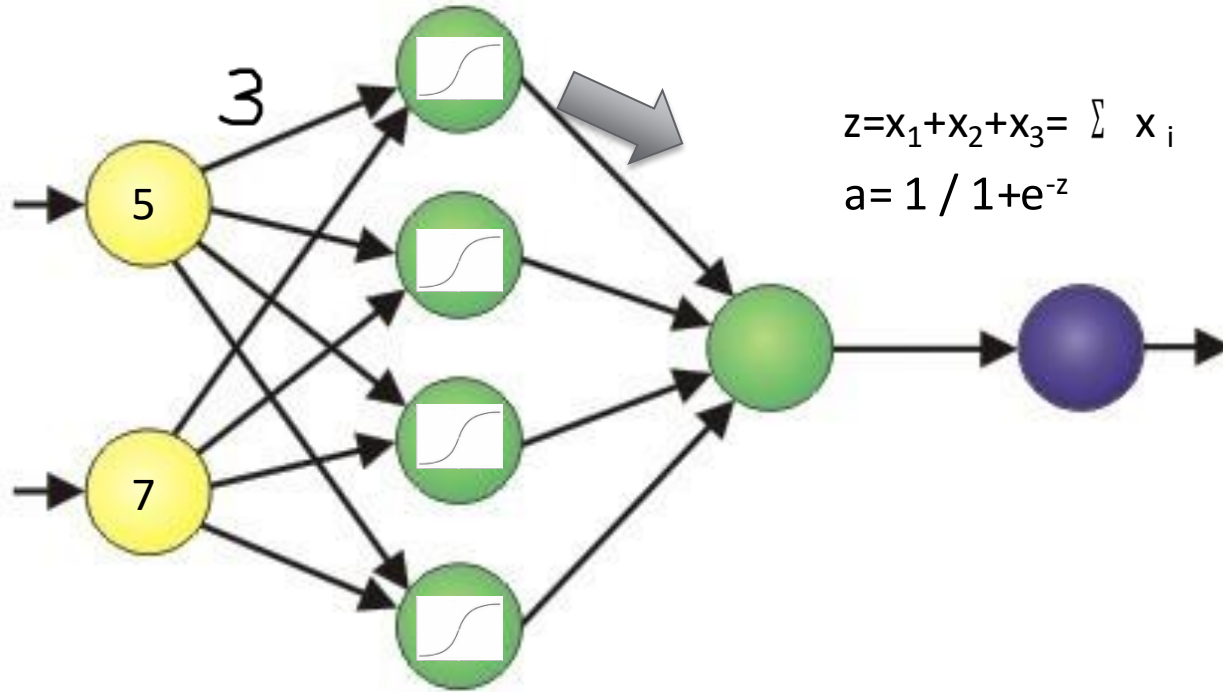
$$Y_{\text{norm}} = y / \max(y)$$



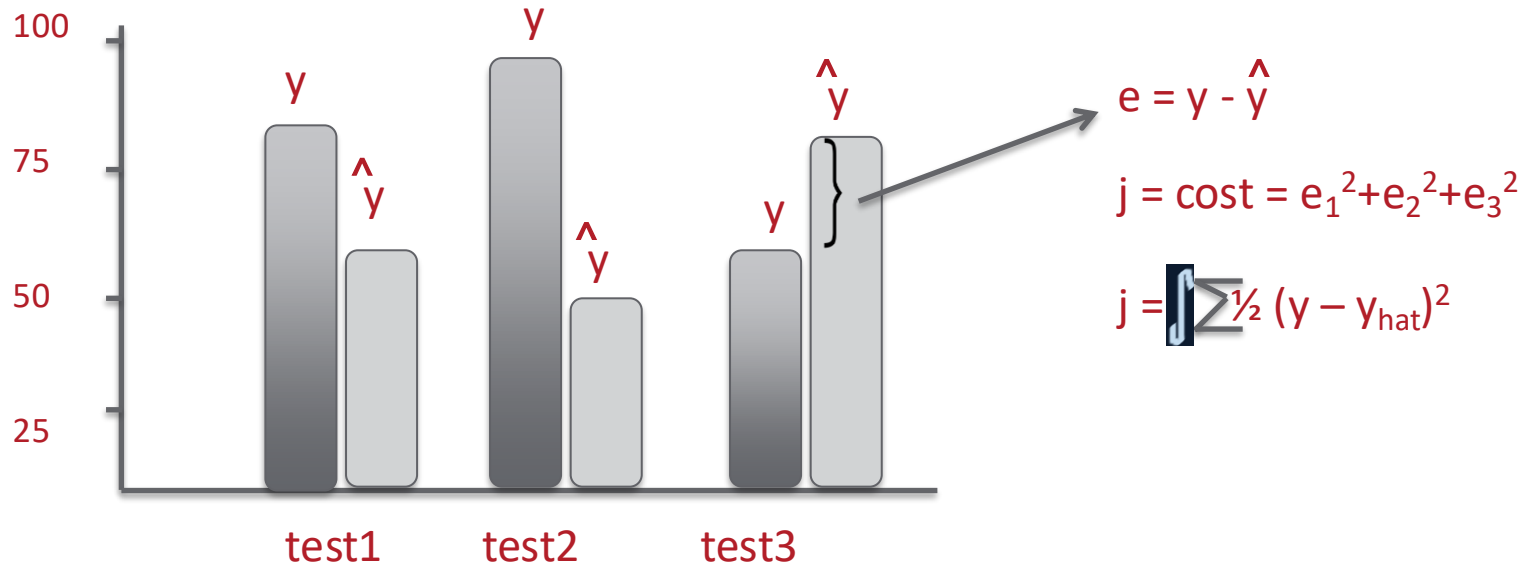
# Basic structure of a neural net



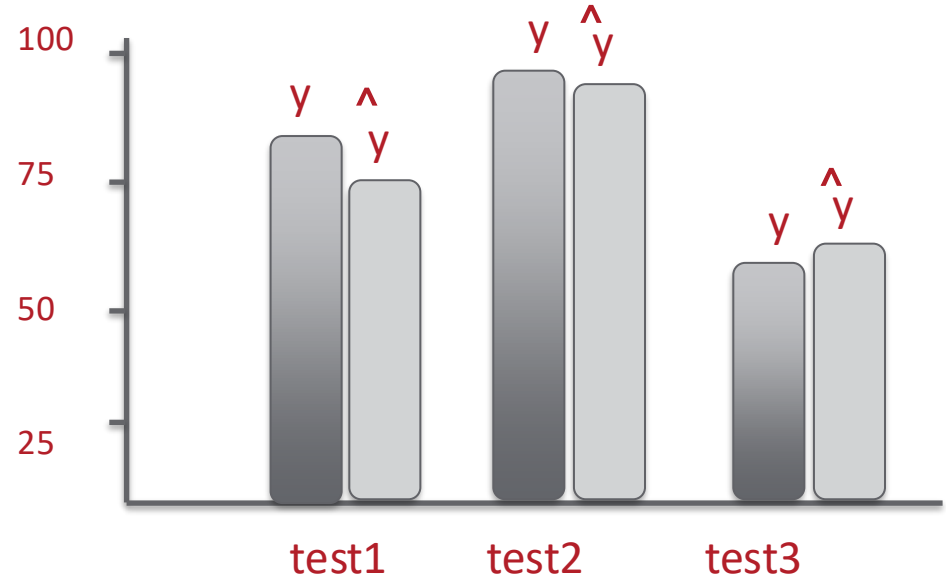
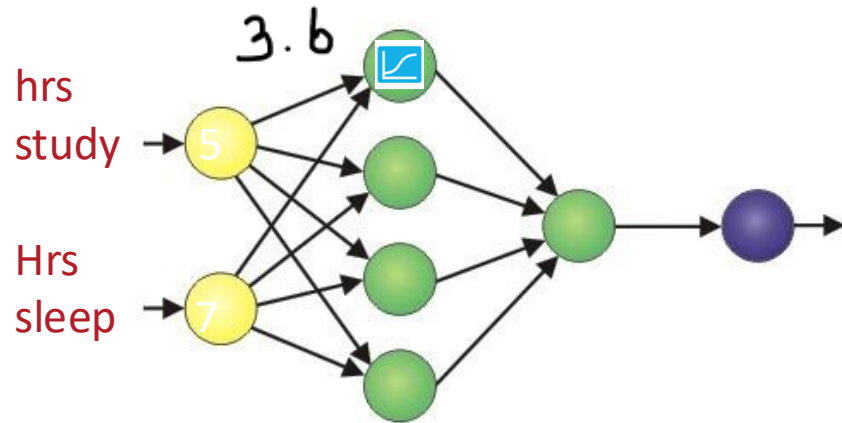
# Sigmoid Operation



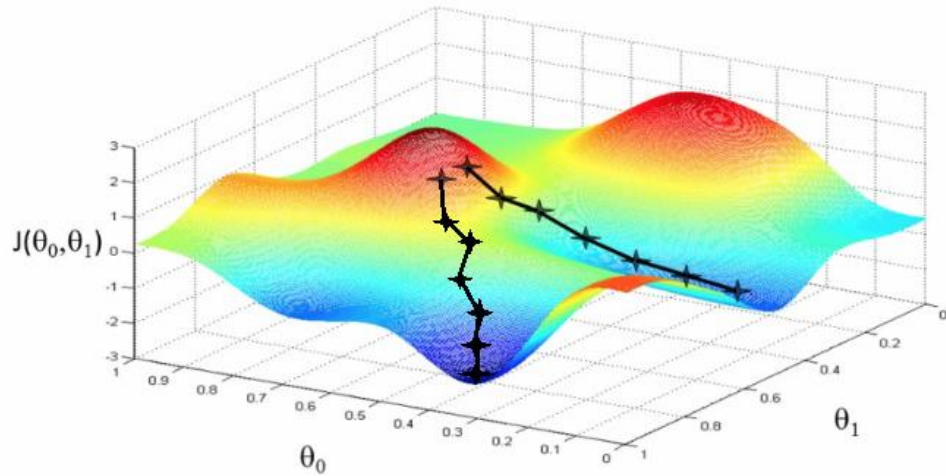
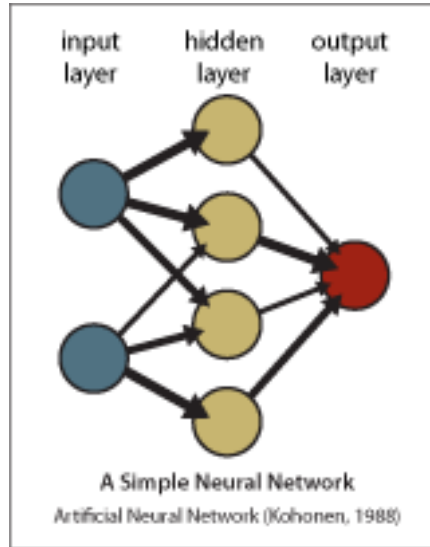
# Training neural network = minimizing cost function



# Backpropagation

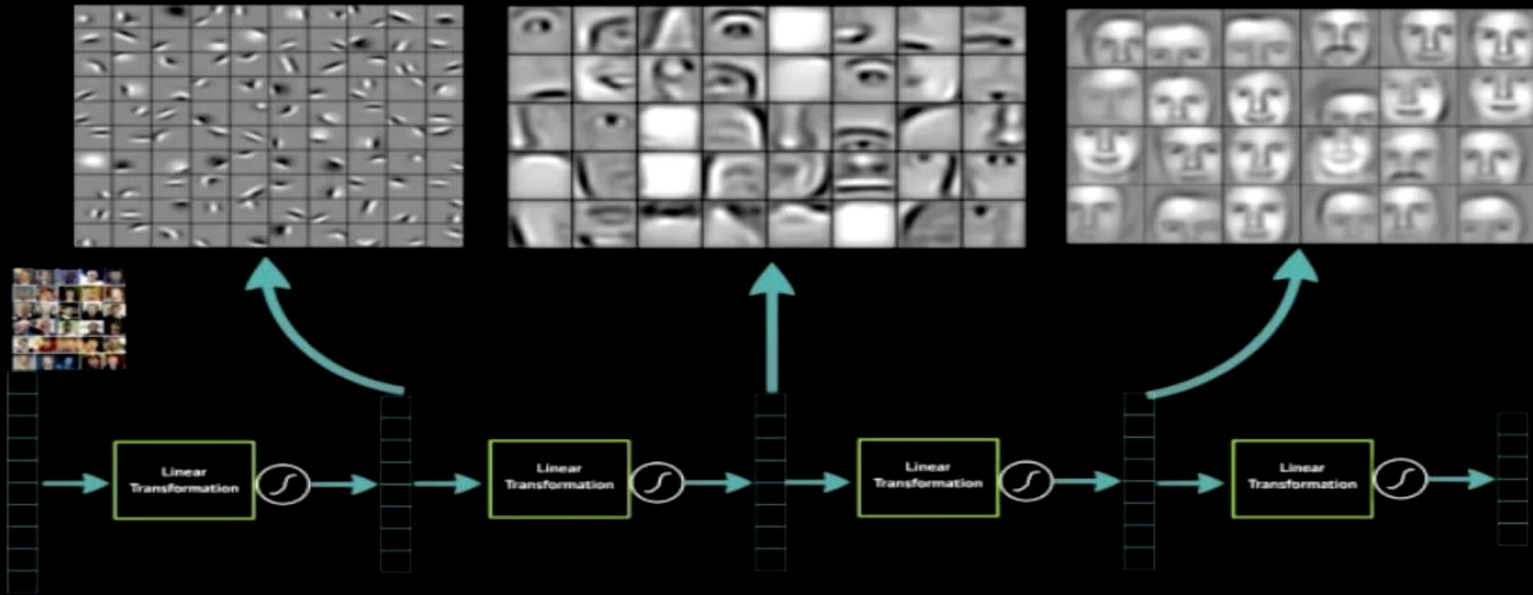


# About neural networks



# Deep learning is about hidden layers (CNN)

## Deep Learning learns layers of features



# Brute Statistics versus Artificial Neural Network

**Regression** takes the data and tries to find the result that minimizes prediction mistakes, maximizing what is called goodness of fit.

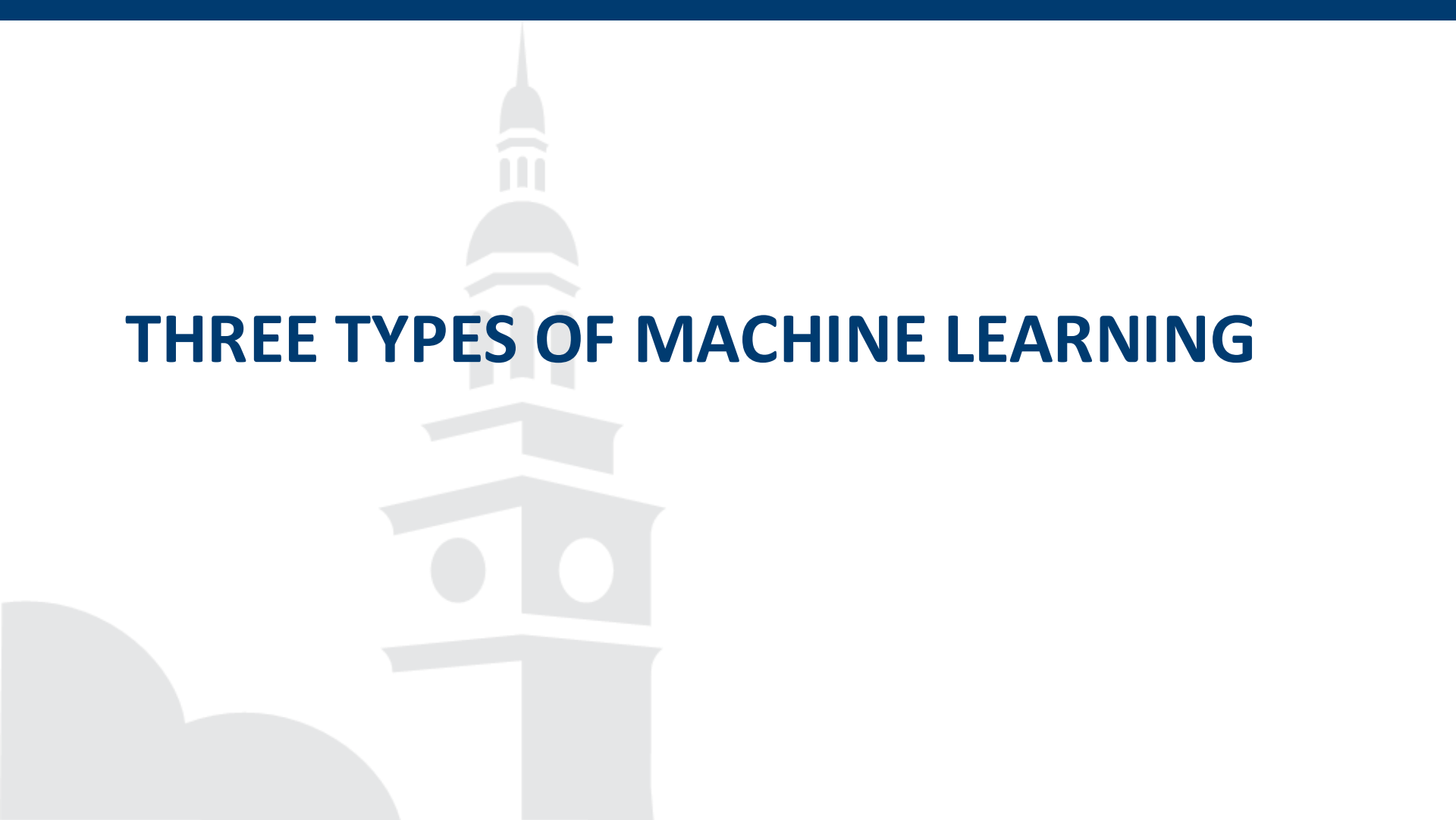
**A physicist, an engineer and a statistician go on a hunting trip....**

Being precisely perfect on average can mean being actually wrong each time. Regression can keep missing several feet to the left or several feet to the right. Even if it averages out to be the correct answer, regression can mean never actually hitting the target.

Unlike regression, **machine learning** predictions might be wrong on average, but when the prediction miss, they often don't miss by much. Statisticians describe this as allowing some bias in exchange for reducing variance.

Inventing a new machine learning method involves proving that it works better in practice. In contrast, inventing a regression method requires first proving that it works in theory, it requires the articulation of a hypothesis.

Machine learning has less need to specify in advance what goes into the model and can accommodate the equivalent of much more complex models with many more interactions between variables.

A faint, light gray background illustration of a building with a dome and a tower, and a cloud. The building has a central dome with a spire, and a tower to the left with two circular windows. A cloud is visible in the bottom left corner.

# THREE TYPES OF MACHINE LEARNING



# Supervised vs. Unsupervised Machine Learning

| Parameters               | Supervised ML techniques                                                                                          | Unsupervised ML techniques                                                                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Process                  | In a supervised learning model, input and output variables are given.                                             | In unsupervised learning model, only input data will be given                                                                             |
| Input data               | Algorithms are trained using labeled data.                                                                        | Algorithms are used against data which is not labeled                                                                                     |
| Algorithms used          | Support vector machine, Neural network, Linear and logistics regression, random forest, and Classification trees. | Unsupervised algorithms can be divided into different categories: like Cluster algorithms, K-means, Hierarchical clustering, etc.         |
| Computational complexity | Supervised learning is a simpler method.                                                                          | Unsupervised learning is computationally complex                                                                                          |
| Use of data              | Supervised learning model uses training data to learn a link between the input and the outputs.                   | Unsupervised learning does not use output data.                                                                                           |
| Accuracy of results      | Highly accurate and trustworthy method.                                                                           | Less accurate and trustworthy method.                                                                                                     |
| Real-time learning       | Learning method takes place offline.                                                                              | Learning method takes place in real time.                                                                                                 |
| Number of classes        | Number of classes is known.                                                                                       | Number of classes is not known.                                                                                                           |
| Main drawback            | Classifying big data can be a real challenge in Supervised Learning.                                              | You cannot get precise information regarding data sorting, and the output as data used in unsupervised learning is labeled and not known. |

# What is reinforcement learning

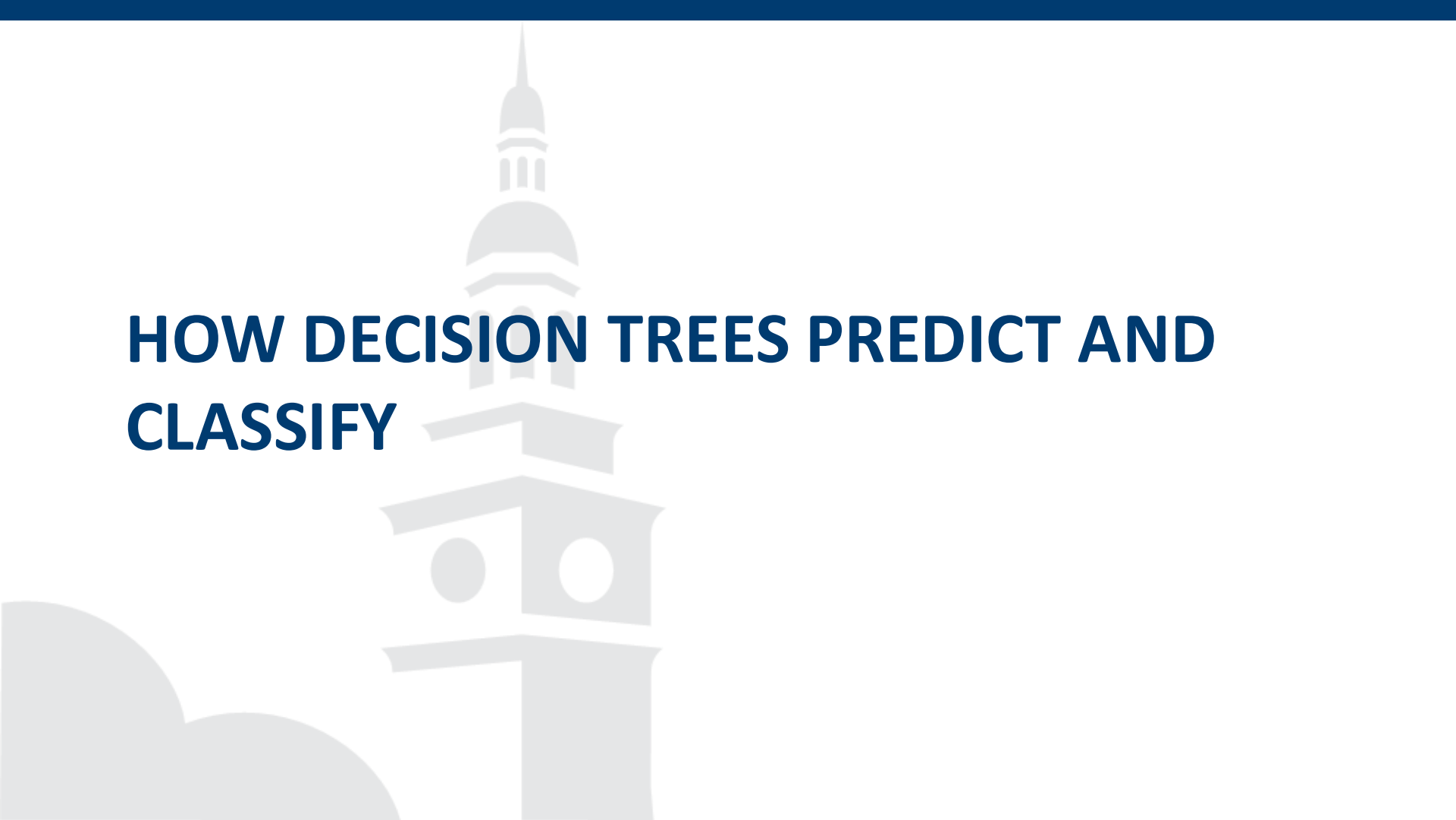
- Reinforcement Learning is a feedback-based Machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions.
- For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.
- In Reinforcement Learning, the agent learns automatically using feedbacks without any labeled data, unlike supervised learning.
- Since there is no labeled data, so the agent is bound to learn by its experience only.



# When do I use reinforcement learning

- RL solves a specific type of problem where decision making is sequential, and the goal is long-term, such as **game-playing, robotics**, etc. The agent interacts with the environment and explores it by itself.
- The primary goal of an agent in reinforcement learning is to improve the performance by getting the maximum positive rewards.
- The agent learns with the process of hit and trial, and based on the experience, it learns to perform the task in a better way. Hence, we can say that ***"Reinforcement learning is a type of machine learning method where an intelligent agent (computer program) interacts with the environment and learns to act within that."*** How a Robotic dog learns the movement of his arms is an example of Reinforcement learning.
- The agent continues doing these three things (**take action, change state/remain in the same state, and get feedback**), and by doing these actions, the agent learns and explores the environment.
- The agent learns that what actions lead to positive feedback or rewards and what actions lead to negative feedback penalty. As a positive reward, the agent gets a positive point, and as a penalty, it gets a negative point.

# HOW DECISION TREES PREDICT AND CLASSIFY

A faint, light gray background illustration of a building with a dome and a tower, and a cloud. The building has a central dome with a spire, and a tower to the right with two circular windows. A cloud is visible in the bottom left corner.

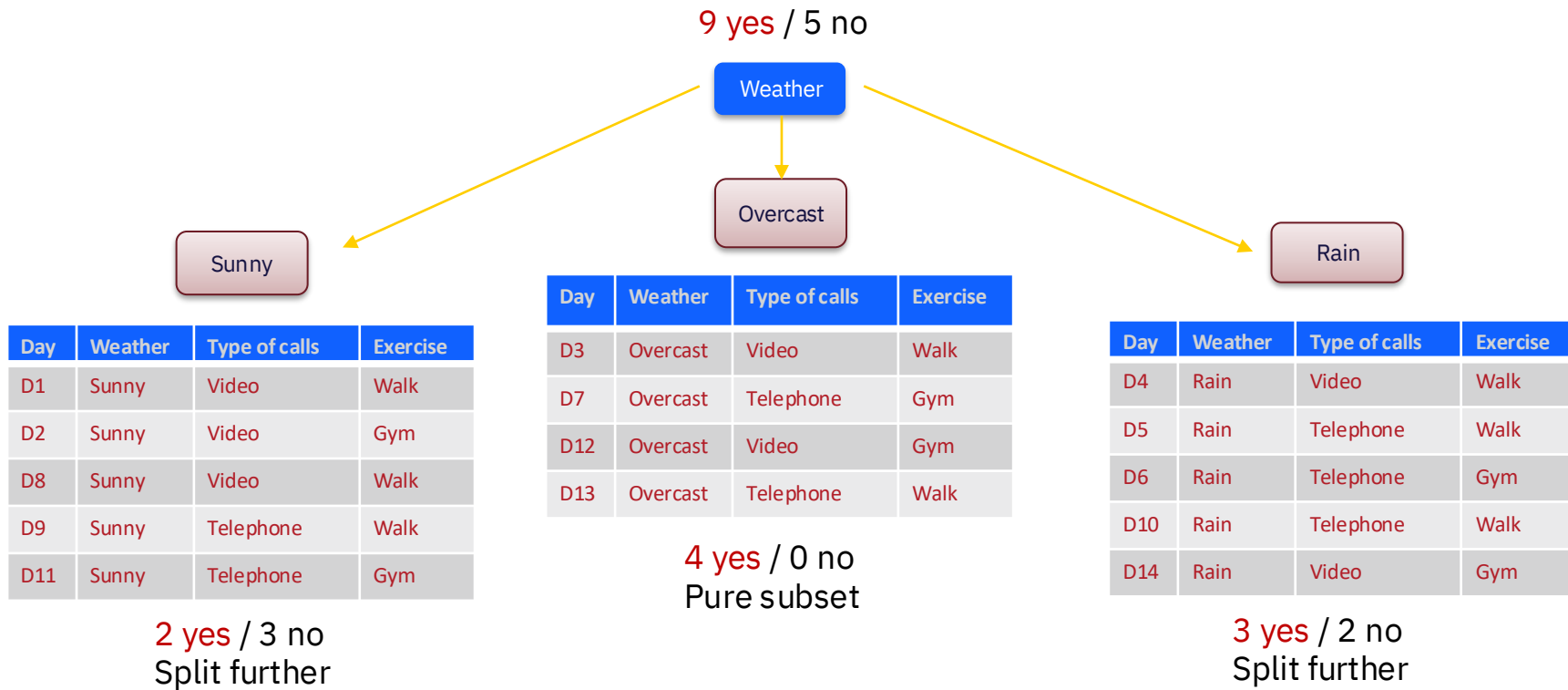
# Predict if Nemra will commute to the office

- **Hard to guess under what conditions Nemra will work from home or drive to the office.**
- **Let's divide and conquer:**
  - Split into subsets
  - Are they all pure (all yes or all no)
  - If yes: stop
  - If no: repeat

| Day | Weather | Type of calls | Exercise | Commute? |
|-----|---------|---------------|----------|----------|
| D15 | Rain    | Video         | Walk     | ?        |

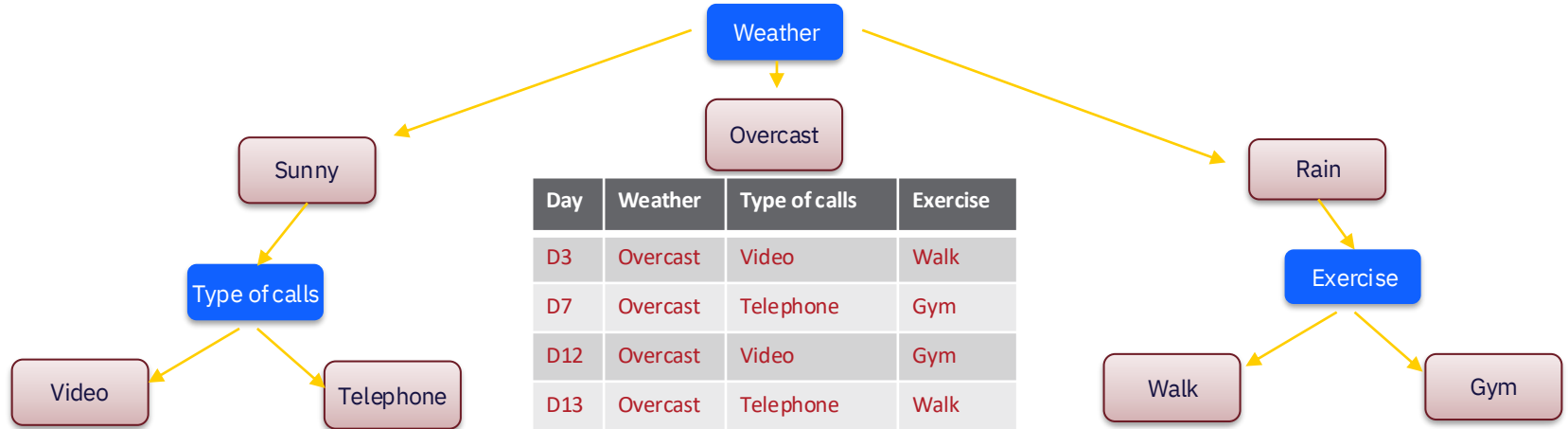
| Day | Weather  | Type of calls    | Exercise | Commutes |
|-----|----------|------------------|----------|----------|
| D1  | Sunny    | Video Conference | Walk     | No       |
| D2  | Sunny    | Video Conference | Gym      | No       |
| D3  | Overcast | Video Conference | Walk     | Yes      |
| D4  | Rain     | Video Conference | Walk     | Yes      |
| D5  | Rain     | Telephone        | Walk     | Yes      |
| D6  | Rain     | Telephone        | Gym      | No       |
| D7  | Overcast | Telephone        | Gym      | Yes      |
| D8  | Sunny    | Video Conference | Walk     | No       |
| D9  | Sunny    | Telephone        | Walk     | Yes      |
| D10 | Rain     | Telephone        | Walk     | Yes      |
| D11 | Sunny    | Telephone        | Gym      | Yes      |
| D12 | Overcast | Video Conference | Gym      | Yes      |
| D13 | Overcast | Telephone        | Walk     | Yes      |
| D14 | Rain     | Video Conference | Gym      | No       |

# The Decision Tree



# If Entropy, then branch further

9 yes / 5 no



| Day | Weather  | Type of calls | Exercise |
|-----|----------|---------------|----------|
| D3  | Overcast | Video         | Walk     |
| D7  | Overcast | Telephone     | Gym      |
| D12 | Overcast | Video         | Gym      |
| D13 | Overcast | Telephone     | Walk     |

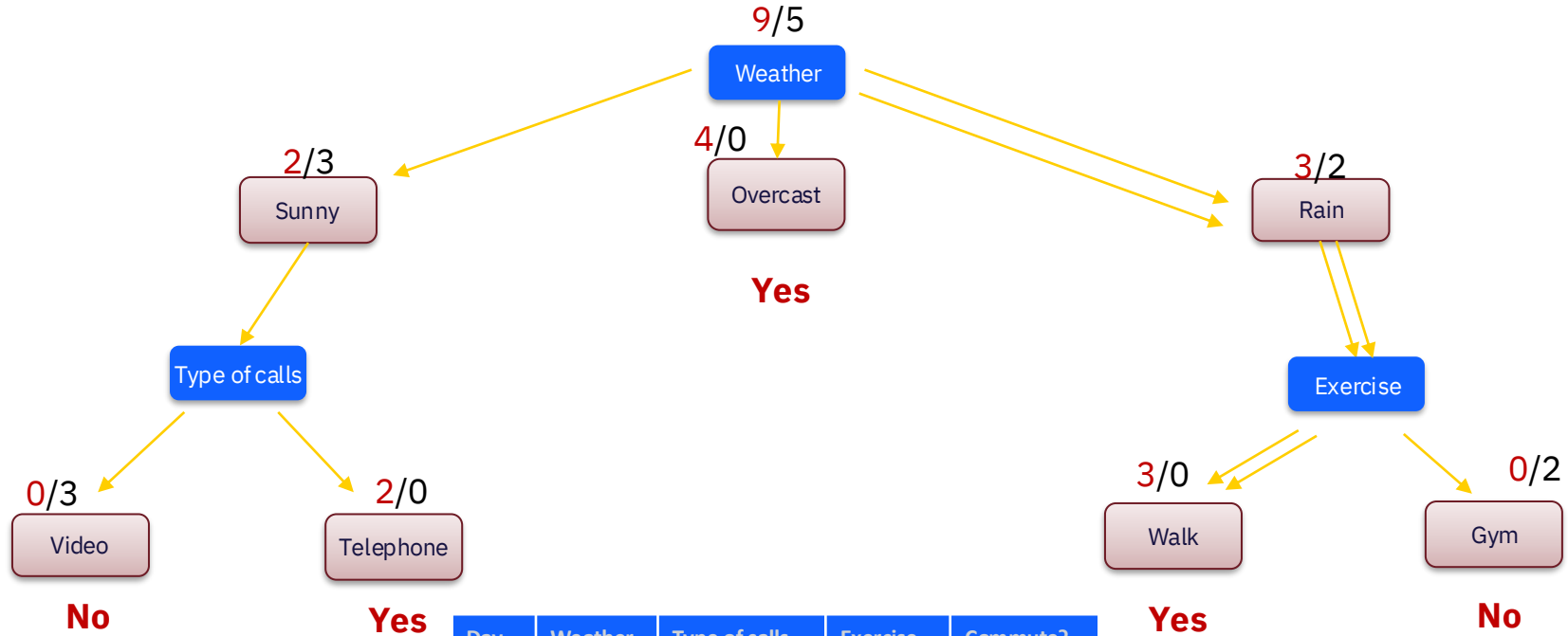
| Day | Type of calls | Exercise |
|-----|---------------|----------|
| D1  | Video         | Walk     |
| D2  | Video         | Gym      |
| D8  | Video         | Walk     |

| Day | Type of calls | Exercise |
|-----|---------------|----------|
| D9  | Telephone     | Walk     |
| D11 | Telephone     | Gym      |

| Day | Type of calls | Exercise |
|-----|---------------|----------|
| D4  | Video         | Walk     |
| D5  | Telephone     | Walk     |
| D10 | Telephone     | Walk     |

| Day | Type of calls | Exercise |
|-----|---------------|----------|
| D6  | Telephone     | Gym      |
| D14 | Video         | Gym      |

# The Prediction

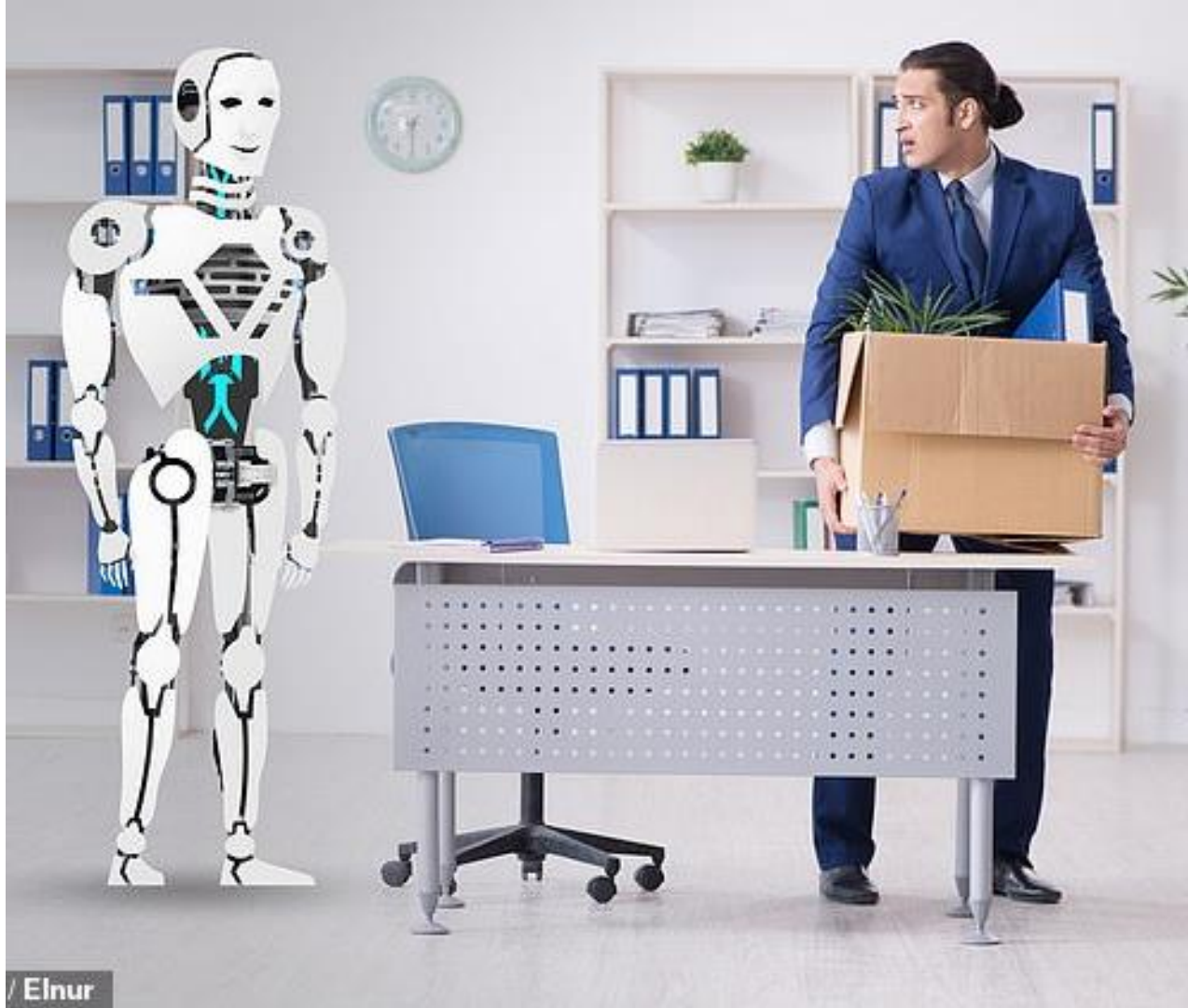


New data: Day 15

| Day | Weather | Type of calls | Exercise | Commute?     |
|-----|---------|---------------|----------|--------------|
| D15 | Rain    | Video         | Walk     | ? <b>Yes</b> |



Is AI  
going to  
Replace  
Me?



## ...not so fast!

Prediction relies on data, that means humans have **two advantages** over machines:

- We know **some things** that machines don't (yet).

- We are better at deciding what to do when **there isn't much data**.

Humans have **three types of data** that machines don't:

- We can gather data from our **senses**: smell, touch, feel, hunch...

- We are the **final arbitrators** of our own preferences. This is why consumer data is extremely valuable. Store discounts for customers who use loyalty cards.

- Privacy concerns** restrict the data available to machines. As long as enough people keep their sexual activity, financial situation, mental health status and repugnant thoughts to themselves, the AI system will have insufficient data to predict behavior.

These excerpts are taken from the book **Prediction Machines** by Ajay Agrawal, Josh Gans and AVI Goldfarb

# THREE MAJOR AI CALIBERS

AI capabilities exist on a spectrum. Here are the three key levels.



## 1 ARTIFICIAL NARROW INTELLIGENCE (ANI)

Sometimes referred to as *Weak AI*

Artificial Narrow Intelligence is AI that specializes in *one* area. ANI can beat Jeopardy! World champion. Ask it to figure out a better way to store data on a hard drive, and it'll look at you blankly.



EXPERT IN  
ONE AREA



## 2 ARTIFICIAL GENERAL INTELLIGENCE (AGI)

Sometimes referred to as *Strong AI*, or *Human-Level AI*

Artificial General Intelligence refers to a computer that is as smart as a human across the board—a machine that can perform any intellectual task that a human being can.



AS SMART AS  
A HUMAN



## 3 ARTIFICIAL SUPERINTELLIGENCE (ASI)

Oxford philosopher and leading AI thinker Nick Bostrom defines superintelligence as “an intellect that is much smarter than the best human brains in practically every field, including scientific creativity, general wisdom and social skills.”

Artificial Superintelligence ranges from a computer that's just a little smarter than a human to one that's trillions of times smarter—across the board.



BEYOND HUMAN  
INTELLIGENCE



FROM SPECIALIZED TO SUPERHUMAN: THE JOURNEY OF INTELLIGENCE. >>>



# THE AI JOURNEY: FROM 2000 TO SUPERINTELLIGENCE

*From learning data to understanding the world to transcending human limits.*



**THE BIG QUESTION:  
HOW WILL WE CHOOSE TO SHAPE THIS FUTURE WITH AI?**

TECHNOLOGY IS COMING.

THE CHOICE IS OURS.

THE FUTURE IS NOT WRITTEN.